

RoPEHAR: A Real-Time Rotary Position Encoding Informer for mmWave-Based Human Activity Recognition in Substations

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Abstract—Safety monitoring of power operations in substations is crucial for accident prevention. However, traditional methods such as wearable devices and video surveillance suffer from limitations including high costs and reliance on lighting conditions. A solution that integrates millimeter-wave radar with deep learning ensures operational compliance by high precision gesture detection. This paper proposes RoPEHAR, a human activity recognition system based on millimeter-wave radar, which combines traditional transformer architecture with rotary positional encoding, specifically designed for human posture recognition in indoor industrial environments. To reduce interference from the coupling of human and instrument signals and noise in electrical scenarios, RoPEHAR introduces a hybrid filtering pipeline that combines hierarchical SNR denoising with enhanced DBSCAN clustering to accurately segment the point cloud data of limbs and instruments. The core innovation lies in the introduction of a spatiotemporal Informer, Roformer. It enhances the 3D-space vector information about data points through dynamic rotary positional encoding. This system effectively models limb motion trajectories. Experiments demonstrate that RoPEHAR achieves high-precision performance, reaching an accuracy of 95.8%, enabling real-time and reliable activity recognition for substation safety monitoring.

Index Terms—Electric power operation safety, human activity recognition, millimeter-wave radar, rotary positional encoding, deep learning.

I. INTRODUCTION

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THE power system is an important infrastructure for global economic development, and the distribution station, as an important link in the power system, plays a key role in transmitting power from the transmission grid to the user terminals [1]. In distribution stations, staff are required to perform a variety of maintenance, inspection and operation tasks to ensure the safe and reliable operation of the power system. However, operation of incorrect instruments and irregularities in the inspection process may lead to personal injuries and economic losses [2].

Traditionally, action recognition of workers has mainly relied on vision-based cameras [3] and infrared array sensors [4]. There are many limitations and challenges associated with this approach. Firstly, distribution substation environments are often characterized by dense electrical equipment, high-voltage components, and confined workspaces. Cameras are constrained by fixed mounting positions and viewing angles, which can easily result in persistent occlusions and missing critical motion cues. Secondly, variations in illumination conditions, reflections from metallic surfaces, and shadows can significantly interfere with reliable pose detection [5], especially during emergency maintenance or power outage scenarios. In addition to vision-based solutions, wearable devices are also commonly used for attitude detection. However, such devices may interfere with normal staff operations and impose additional costs [6]. More importantly, in high-voltage substations, body-worn sensors may raise safety compliance concerns and reduce worker comfort when combined with mandatory protective equipment, limiting their long-term practicality.

In recent years, with the rapid development of wireless sensing related work [7], continuous frequency modulated wave (FMCW) radio technology has been widely used in many scenarios. Examples include vital signs [8], automotive radar [9], and human bone reconstruction [10], [11]. Millimeter wave radar is a technology that uses electromagnetic waves in the millimeter wave band (30-300GHz) for detection, which emits FMCW signals and receives their reflected signals to obtain target information. This makes it is not sensitive to ambient light. The short wavelength characteristic gives millimeter wave radar a strong penetration capability with high resolution. Therefore, the use of millimeter wave radar for HAR is considered a promising solution.

Millimeter-wave radar for staff position detection under power system monitoring faces some special challenges [12].

Current research efforts are mainly limited to the laboratory principle prototype stage, while applications in complex environments such as power distribution stations have not been extensively explored. Compared with the general behaviour in daily life, power workers need to face more complex behaviours. In addition, due to the special nature of electric power work, there is close contact between staff and equipment, which makes it difficult to accurately differentiate between signal echoes from the human body and equipment.

To address this, we propose RoPEHAR, a personnel action detection system for power distribution stations based on the IWR1843 mmWave radar sensor. The mmWave radar sensor wirelessly transmits human motion point cloud data, enabling easy deployment in power station environments. To mitigate signal coupling between human bodies and instruments, RoPEHAR introduces a hybrid filtering pipeline to precisely segment instrument and human data.

To overcome the limitations of single-model approaches and enhance the extraction of limb movement trajectory features, this paper proposes a spatiotemporal Informer named Roformer with Rotary Position Encoding (RoPE). The RoPE directly models local relative positional relationships in action sequences through vector rotation, allowing the model to accurately perceive spatial distance variations between key joints in limb motion trajectories.

Furthermore, the study employs environmental voxelization and dimensionality reduction techniques to significantly enhance computational efficiency while preserving critical motion features, enabling the model to meet stringent real-time response requirements for action monitoring. The proposed method achieves high accuracy in human pose recognition and holds promising potential to provide robust support for the operation and management of power distribution stations.

We summarize the contributions of this work as follows:

- Proposed RoPEHAR, the first mmWave radar-based HAR system tailored for substations. The framework integrates tailored voxelization and dimensionality reduction to preserve key interaction features, thereby enabling real-time action monitoring under the stringent latency constraints of power operation safety.
- Developed a hybrid denoising pipeline that integrates hierarchical filtering based on signal-to-noise ratio (SNR) with weighted Density-Based Spatial Clustering of Applications with Noise (DBSCAN), achieving multi-threshold separation of human and equipment echoes in substations. This design effectively suppresses electromagnetic interference (EMI) and multipath noise, enabling accurate extraction of limb trajectories under complex electrical environments.
- Introduced Roformer, an Informer-based model enhanced with RoPE. By modeling relative spatial distances between worker limbs and electrical apparatus, RoPE effectively captures fine-grained angular dependencies and tool-centric motions, thereby improving robustness against overlapping human–equipment reflections.

II. RELATED WORK

A. Power System Monitoring

Research on personnel monitoring has been conducted for many years [13], with wearable device-based solutions being among the earliest and most established approaches. Awolusi et al. [14] provided a comprehensive review of wearable technologies for industrial safety monitoring, highlighting the use of accelerometers, gyroscopes, and smart helmets for tracking worker movements. Ometov et al. [15] further emphasized limitations related to battery life, sensor reliability, and risks associated with EMI and potential arc flash hazards in industrial environments.

In contrast, vision-based monitoring technologies offer a more convenient alternative. Peng et al. [16] developed CoryNet, a deep learning framework for real-time detection of safety violations using video surveillance. Bo et al. [17] proposed a skeleton-based action recognition method using graph convolutional networks for safety supervision in power distribution scenarios, demonstrating the potential of pose estimation techniques for identifying unsafe behaviors. However, substations are characterized by dense equipment layouts, where electrical devices frequently occlude workers' limbs. In addition, indoor distribution rooms rely heavily on artificial lighting, and emergency maintenance during power outages disrupts natural illumination, rendering vision-based monitoring ineffective.

More recently, PowerSkel [18], a power system monitoring scheme based on channel state information signals, has been proposed to address light insensitivity and human posture estimation without requiring additional wearable devices. Nevertheless, CSI-based methods depend on dense Wi-Fi access point deployment and are highly sensitive to environmental changes. In substations where equipment configurations frequently vary due to upgrades or maintenance, such systems require repeated recalibration.

Although millimeter-wave radar-based human activity recognition (HAR) has shown promising results in controlled laboratory environments [19], most existing radar HAR systems are validated in empty rooms with minimal interference. In real substations, severe multipath propagation and EMI [20], together with close human–equipment interactions, result in coupled limb–equipment radar echoes that challenge existing signal processing approaches.

B. Point Cloud-based Pose Recognition

In the field of 3D point cloud data processing, PointNet [21] and PointNet++ [22] laid critical foundations for raw point cloud feature learning through pointwise extraction and hierarchical multi-scale perception, respectively. To address frame-wise point cloud quantity imbalance, Akash Deep Singh et al. proposed RadHAR, which leverages sliding time windows for point cloud accumulation and achieved five-class action recognition using a CNN+bidirectional LSTM classifier [23]. Building on this, Chengxi Yu et al. developed the non-invasive radar sensing system DVCNN [24], which generates spatiotemporal point cloud features through enhanced voxelization and improves robustness via data augmentation and

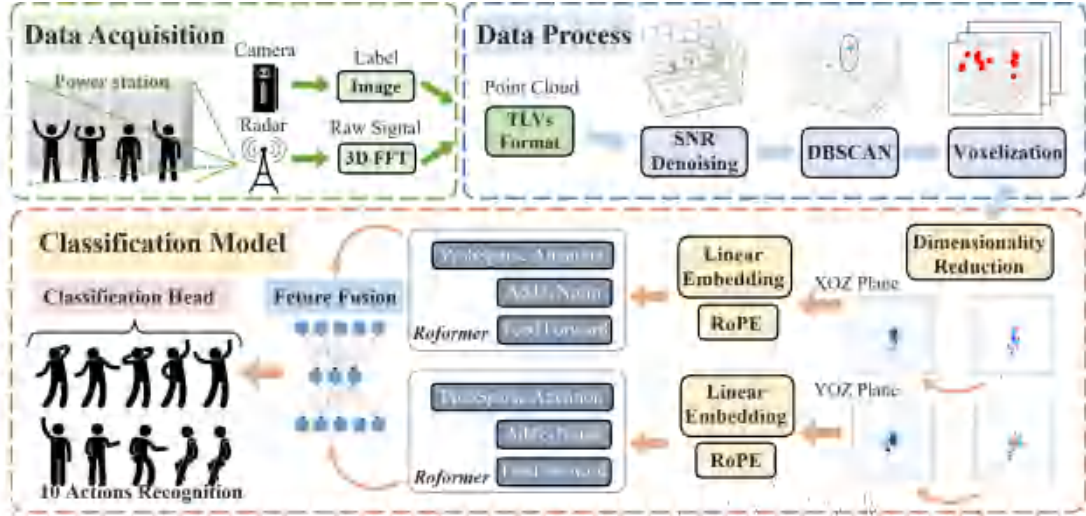


Fig. 1. The framework of RoPEHAR. RoPEHAR establishes communication by packaging raw radar data into TLV packets and transmitting them to the host computer. A Python script is used for preprocessing the point cloud data. Camera data are used solely for offline action category label generation during dataset construction and are not involved in Roformer’s training or inference process.

dual-view machine learning. To further address point cloud sparsity, Hao et al. proposed mm-ARnet [25], demonstrating that multi-frame fusion with stochastic sampling effectively enhances point cloud density while preserving temporal structure, supporting temporal aggregation as a practical strategy for sparse mmWave HAR.

Beyond sparsity, data acquisition itself remains a fundamental bottleneck. Video2mmPoint [26] synthesizes mmWave point clouds from monocular videos through skeleton-matching and statistical synthesis, improving gait recognition when real radar data are limited, though camera dependency constrains low-light deployment. Alternatively, simulation-based domain adaptation has been explored to generate synthetic FMCW radar data aligned to real-world distributions [27], reducing collection overhead, yet such frameworks are typically validated in controlled geometries and may not capture the complex EMI conditions of real substations. MM-HAT [28] pioneered Transformer-based mmWave HAR by integrating raw point clouds with Kalman-filtered target data to handle signal instability. However, this method lacks explicit relative spatial encoding mechanisms critical for fine-grained industrial operations.

C. Deep Learning Based Classification for Action Recognition

Action recognition tasks involve capturing complex spatiotemporal dynamics, where identical actions may exhibit significant variations under different conditions. Traditional methods rely on handcrafted feature extractors such as optical flow, HOG, and SIFT to encode environmental robustness [29]. However, these approaches typically process spatial and temporal information separately, failing to model their nonlinear interactions.

Deep learning has transformed this field. Convolutional neural networks (CNNs) automatically extract high-level features from raw data, minimizing manual intervention [30]. Using multi-layer architectures, nonlinear activations, and parameter optimization, these models capture spatio-temporal

dependencies in videos, modeling complex interactions between motion dynamics and temporal sequences [31]. This technological trend has further extended into point cloud video analysis and millimeter-wave radar sensing. For instance, the P4Transformer [32] proposed by Hehe Fan et al. integrates 4D spatio-temporal convolution with a Transformer-based self-attention mechanism to directly model local structures and global motion in raw point cloud videos, eliminating the need for explicit point tracking. Despite these advances, a common bottleneck across activity recognition approaches is the reliance on large-scale annotated datasets, which are particularly costly to obtain in restricted industrial environments. To address this, active learning with informative sample selection has been explored to identify the most valuable samples for annotation under limited training data [33], offering a promising direction for reducing annotation overhead in data-scarce industrial HAR scenarios such as substations.

Recently, RadarLLM [34] integrates large language models with mmWave radar for semantic motion understanding via cross-modal alignment, but its reliance on large-scale paired data and heavy computation makes it unsuitable for real-time industrial deployment.

III. METHOD

A. RoPEHAR System Overview

The RoPEHAR framework comprises three core components: data acquisition, data processing, and classification models, as illustrated in Fig. 1. During the data acquisition phase, multiple distribution cabinets and a millimeter-wave radar system are deployed in a real-world substation. The radar captures raw FMCW signals from electrical workers, which are converted into 3D point clouds through a 3D-FFT processing chain and packaged into Type-Length-Value (TLV) format for transmission to the server. In the data processing stage, a hybrid denoising pipeline integrating hierarchical SNR-based filtering and enhanced DBSCAN clustering is applied to

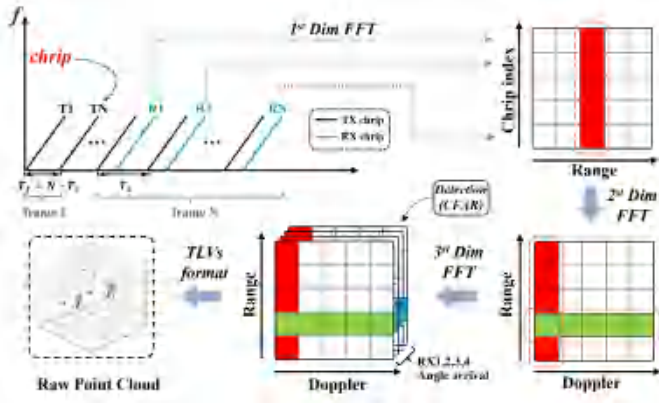


Fig. 2. 3D-FFT radar full signal processing chain.

extract critical human behavioral features from raw point cloud data. Processed human motion data then undergoes specific voxelization and dimensionality reduction, transforming sparse point clouds into standardized 2D projections. Finally, the Reformer classification model recognizes 10 power operation actions by analyzing these projections.

B. Data Acquisition

In radar systems, echoes from targets are inherently modulated by parameters such as distance, velocity, and angle. The engineering approach employs a cascaded FFT framework (3D-FFT) that systematically resolves overlapping target signatures across three orthogonal dimensions. The computation begins with raw chirp signals captured by IWR1843BOOST. As shown in Fig. 2, the operational workflow of each stage is as follows:

- 1) Range Detection (1st FFT): FMCW chirp signals are transmitted, and each echo signal undergoes time-frequency conversion via FFT, generating a single-cycle range profile.
- 2) Velocity Estimation (2nd FFT): A cross-chirp FFT is performed on range data from consecutive chirps to analyze Doppler-induced phase variations.
- 3) Angle-of-Arrival Calculation (3rd FFT): Spatial-dimension FFT is applied to range-Doppler data from multiple receive antennas, ultimately constructing a 3D range-velocity-angle localization map.
- 4) Constant False Alarm Rate(CFAR): To suppress random noise and highlight valid targets, CFAR thresholding scheme is applied on the 3D map. By adaptively adjusting detection thresholds according to local noise statistics, CFAR enhances robustness against clutter.

C. Data Process

Unlike data acquisition in traditional empty laboratory settings, power distribution rooms are characterized by EMI, complex geometric layouts, and occlusion artifacts from densely arranged equipment, making it challenging to extract critical human motion data. To overcome these issues, we propose a hybrid denoising pipeline that integrates hierarchical SNR filtering and weighted DBSCAN clustering algorithm.

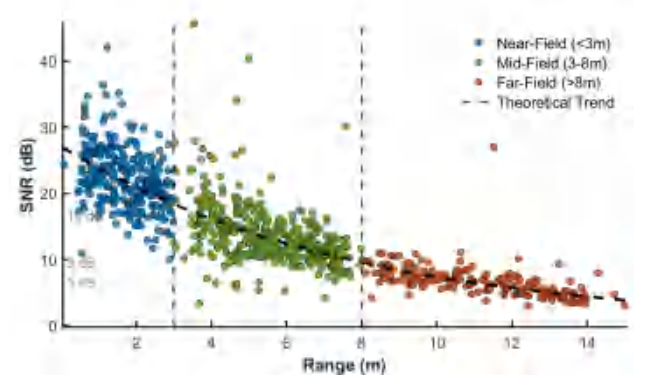


Fig. 3. Multi-Region SNR-Range Relationship with Hierarchical Multiplicative Linear Model. The blue regions represent human point clouds, green regions denote distribution cabinets, and orange regions correspond to random noise.

1) *Hierarchical SNR Based Denoising*: Given the high sensitivity of the millimeter-wave frequency range, radar can detect small-scale variations in the environment, leading to the random occurrence of noise in the point cloud generated by radar signals during data collection [35]. According to the method in reference [36], the measurement range of the sensor is divided into three regions, and the observed values within each region are filtered using different SNR thresholds through the hierarchical multiplicative linear model:

$$\log(SNR_i) = \beta_0 + \beta_1 \times Range_i + \xi_i \quad (1)$$

where β_i is the regression slope, ξ is the random unexplained error, and $\xi \sim \text{iid}N(0, \sigma^2)$.

We determine the range of the threshold based on the lower limit value in the confidence interval. The three different regions are divided at 3m, 8m, and 15m. Fig. 3 shows the range of these thresholds and the corresponding SNR distributions for different objects in the substation environment.

Through the hierarchical multiplicative linear model (Eq. (1)), it is established that point cloud data corresponding to different distances and objects exhibit distinct noise signals and SNR echoes. A clear trend is observed: the higher the SNR of a point, the higher the certainty that the point corresponds to a person. This hierarchical SNR denoising strategy adaptively adjusts thresholds based on distance and angle, better preserving human body structure compared to conventional global fixed-threshold filtering, as demonstrated in Fig. 4 (b-c). However, hierarchical SNR denoising alone still retains some static equipment points irrelevant to human actions, lacking spatial semantic filtering capability.

2) *Enhanced Density-Based Spatial Clustering of Applications with Noise*: DBSCAN is a density-based clustering algorithm that groups data points according to local spatial density while explicitly identifying noise [37]. The clustering behavior is governed by two key parameters: the neighborhood radius ϵ and the minimum number of points $MinPts$, which together determine whether a point belongs to a dense region or is treated as an outlier. In this work, these parameters are selected through multi-stage experimental validation to balance noise suppression and cluster integrity under partial

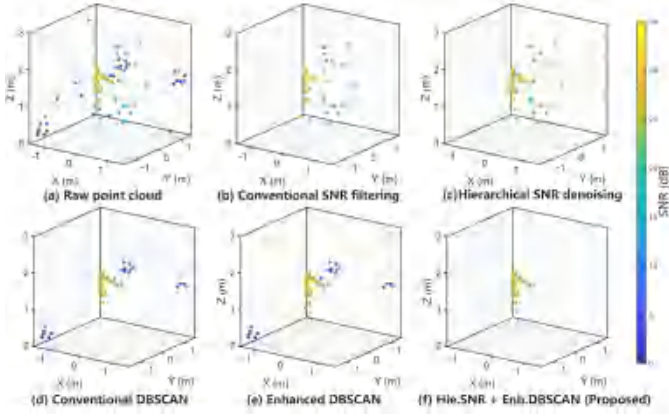


Fig. 4. Comparative visualization of point cloud denoising methods in a real substation scenario.

occlusion. The neighborhood radius ϵ is optimized via grid search, while $MinPts$ is chosen to effectively suppress multipath noise without fragmenting human clusters. Based on these evaluations, we set $\epsilon = 0.04$ and $MinPts = 40$, and the cluster with the largest number of points is selected as the representative human point cloud.

Nevertheless, conventional DBSCAN with standard Euclidean distance faces challenges in substation environments. Due to the similar density and spatial overlap of human and equipment point clouds, conventional DBSCAN struggles to effectively distinguish between human and equipment clusters, and tends to cluster only the upper body while erroneously filtering out critical parts such as legs and hands, as shown in Fig. 4 (d). This is because points from the same individual generally exhibit coherence in the horizontal plane, whereas they tend to be more dispersed along the vertical axis.

To mitigate this bias, a modified Euclidean distance metric is adopted by assigning lower weights to the vertical component during clustering. The weighted distance is computed as:

$$d_{\text{weighted}} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + \alpha^2(z_1 - z_2)^2} \quad (2)$$

where α is the vertical weighting factor. Conventional DBSCAN uses $\alpha = 1$, while our enhanced DBSCAN adopts $\alpha = 0.4$ to account for the anisotropic distribution of human body points. This modification substantially improves the completeness of the extracted human point cloud, as illustrated in Fig. 4 (e), although it remains susceptible to dense environmental noise when applied in isolation.

The proposed hybrid approach, shown in Fig. 4 (f), integrates hierarchical SNR denoising with enhanced DBSCAN, first suppressing environmental noise and weak echoes through adaptive SNR thresholds, then further filtering human-related point clouds and removing isolated equipment clusters using enhanced DBSCAN. This hybrid approach maximally preserves the complete human body structure while significantly enhancing the semantic purity of the point cloud. The overall hybrid filtering framework, which integrates SNR-based denoising with density-based clustering, is summarized in Table I.

3) *Environmental voxelization*: Traditional methods will generate homogeneous voxels data, but they cannot capture

TABLE I
HYBRID SNR-DBSCAN FILTERING PROCESS

Algorithm 1 Hybrid SNR-DBSCAN

Input: Raw point cloud data, SNR thresholds T_i , neighborhood radius ϵ , $MinPts$, weight α

Output: Human point cloud clusters

- 1: Initialise cluster counter $C = 0$;
- 2: **for** each unvisited point P in Data **do**
- 3: **if** $SNR(P) < T_i$ **then** mark P as Noise and **continue**;
- 4: Find ϵ -neighbors of P using weighted distance:

$$D(p_i, p_j) = \sqrt{(p_i^x - p_j^x)^2 + (p_i^y - p_j^y)^2 + \alpha^2(p_i^z - p_j^z)^2};$$
- 5: Remove neighbors with $SNR < T_i$;
- 6: **if** $|NeighborPts| < MinPts$ **then** mark P as Noise;
- 7: **else** set $C = C + 1$, add P and neighbors to cluster C ;
- 8: Expand cluster by recursively visiting unassigned neighbors;
- 9: **end for**

space background information in physical environment [24]. To solve this problem, we capture 3D-space voxels figures including their environment. The procedure is stated as follows. Suppose that there exist n points in a frame and the coordinates of the i th point are denoted by (x_i, y_i, z_i) . Thus, the length R_l , width R_w , and height R_h of the cube confining the voxels can be determined as:

$$x_{\text{res}} \stackrel{\text{def}}{=} R_l / V_l \quad (3)$$

$$y_{\text{res}} \stackrel{\text{def}}{=} R_w / V_w \quad (4)$$

$$z_{\text{res}} \stackrel{\text{def}}{=} R_h / V_h \quad (5)$$

Assuming a point cloud file contains n points, where the coordinates of the i -th point are denoted as (x_i, y_i, z_i) , the voxel coordinates corresponding to the x , y , and z axes are represented as x_{ind_i} , y_{ind_i} , and z_{ind_i} , respectively. The voxel grid matching for each point is determined based on the following formulas, considering the minimum values $(x_{\min}, y_{\min}, z_{\min})$ for each corresponding axis:

$$x_{\text{ind}_i} \stackrel{\text{def}}{=} (x_i - x_{\min}) / x_{\text{res}} \quad (6)$$

$$y_{\text{ind}_i} \stackrel{\text{def}}{=} (y_i - y_{\min}) / y_{\text{res}} \quad (7)$$

$$z_{\text{ind}_i} \stackrel{\text{def}}{=} (z_i - z_{\min}) / z_{\text{res}} \quad (8)$$

The above-mentioned method should be modified based on the actual size of the power distribution room and the sensing range of the radar to adjust the voxel size. Therefore, for a background environment of a $10\text{m} \times 6\text{m} \times 10\text{m}$ power distribution room, we have set up a voxel grid of $25 \times 15 \times 25$. It is worth noting that this fixed-grid discretization normalizes the spatial distribution of radar reflections into a standardized coordinate space, thereby reducing sensitivity to gross anthropometric differences such as subject height and build.

D. Classification Model

1) *Dimension Reduction*: Considering the significantly large input size and the extreme sparsity of the voxel space, we used a triangulation method to project the voxel grid

from the front and side views onto 2D images. We projected the reflection points onto the depth-azimuth (XOZ) and depth-elevation (YOZ) planes separately. Subsequently, we constructed grayscale images of size 15×25 and 25×25 . Each pixel in the image corresponds to a voxel grid in the voxel space, and the pixel channels represent different voxel data projections depending on the corresponding plane. Pixels that were not detected were assigned a value of 0. This preprocessing significantly reduced the computational complexity of the network.

2) *Probabilistic Attention of Informer*: Traditional HAR methods utilizing millimeter-wave radar data, such as LSTM and SVM, demonstrate severe limitations in complex industrial settings. Electrical operations involve sophisticated spatio-temporal interactions between intricate limb trajectories and instrument positioning. Constrained by sequential processing architectures, conventional models fail to establish long-range dependencies and capture non-linear spatio-temporal correlations. In contrast, Transformer's permutation-invariant attention mechanism adaptively concentrates on salient spatial features while suppressing noise interference from electrical apparatus.

However, HAR in industrial substations requires not only accurate spatiotemporal modeling but also strict real-time performance to ensure safety monitoring and operator guidance. Conventional Transformer architectures employ Multi-Head Self-Attention (MHSA), where the computation of attention scores across all query-key pairs leads to quadratic time complexity $O(L^2)$. Such computational demands become prohibitive when processing long-sequence radar data in real-time scenarios, resulting in latency that hinders immediate activity classification and response.

To overcome this challenge, we adopt the Informer architecture [38], which introduces ProbSparse Attention and reduces the time complexity to $O(L \log L)$. The mechanism achieves this by sparsely sampling query-key pairs: instead of exhaustively computing all attention scores, the model first evaluates the contribution of each key, selects the most informative subsets, and then focuses attention computation only on these high-contribution pairs.

In this Informer framework, the input is typically split into three components: query, key, and value. We then proceed to compute the similarity between the query and the key, followed by a weighted aggregation of the associated values using the following formula:

$$A(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V \quad (9)$$

where $Q \in \mathbb{R}^{L_Q \times d}$, $K \in \mathbb{R}^{L_K \times d}$, $V \in \mathbb{R}^{L_V \times d}$. For further elaboration, we let q_i , k_j , and v_i represent Q , K , and V , respectively. We calculate the inner product of each element $a_{ij} = q_i k_j^T$ with all the others, which is the reason why the complexity is $O(L^2)$. The idea of ProbAttention integrates two additional computation steps into the attenuation process. Firstly, the model randomly chooses k_j and subsequently computes its inner product with all other elements. Secondly, following the previous step's outcome, the model selects $\ln(L'_q)$, q_i and internally multiplies them with all k_j to

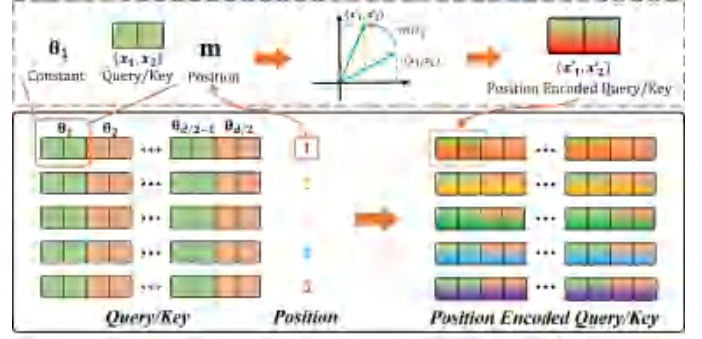


Fig. 5. The implementation of rotary position encoding.

obtain the attention fractional moments, which produces the attention score matrix. The above two steps' total algorithmic complexity stands at $O(L \log L)$.

3) *Rotary Position Encoding*: Traditional Informer architectures adopt absolute positional encoding, where each token in the sequence is assigned a fixed position vector that represents its absolute location. While this method is effective for generic sequential tasks such as natural language processing, it exhibits significant limitations when applied to HAR in industrial substations. In particular, electrical operations often involve large-amplitude, irregular limb trajectories as technicians interact with instruments under different working postures. Since absolute encoding lacks the ability to explicitly represent relative positional dependencies between tokens, it fails to model the subtle correlations that arise from continuous, dynamic body movements in 3D space.

To address this challenge, we employ RoPE to enhance the temporal-spatial representation of radar point cloud sequences. Unlike static absolute encoding, RoPE introduces positional information by rotating the embedding vectors of tokens based on their relative distances. This enables the model to capture both local motion continuity and long-range dependencies. In essence, RoPE provides a mechanism to dynamically align feature representations with the actual kinematic structure of human actions in electrical tasks.

Specifically, for each token in the sequence, its position vector is rotated by an angle determined by its relative offset to other tokens. This results in a new set of rotated position vectors that are fused with the original token embeddings [39], as illustrated in Fig. 5. By embedding relative geometry directly into the query-key interactions, the Informer can better discriminate between fine-grained limb trajectories while simultaneously filtering out irrelevant background fluctuations caused by electrical equipment.

Formally, for any q_i , k_j , a RoPE $R_{\theta, pos}^d$ is multiplied:

$$rq_i = R_{\theta, i}^d q_i, \quad rk_j = R_{\theta, j}^d k_j \quad (10)$$

The inner product with the addition of rotational position coding can be expressed as follows:

$$\begin{aligned} rq_i rk_j^T &= (R_{\theta, i}^d q_i) (R_{\theta, j}^d k_j)^T \\ &= q_i R_{\theta, j-i}^d k_j^T \end{aligned} \quad (11)$$

The above formula accurately represents the relative distance between positions i and j . Its remarkable feature lies in

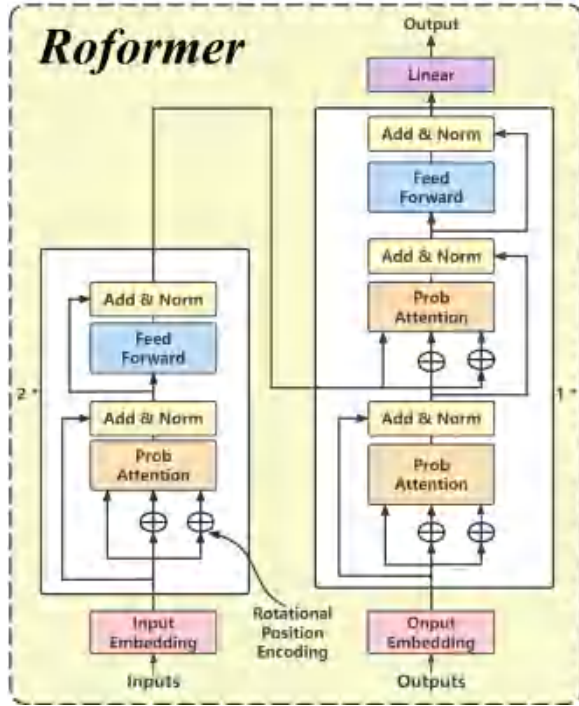


Fig. 6. Roformer system based on Informer and rotary position encoding.

the rotations of the sine-cosine matrix. The rotary position embedding matrix is defined as:

$$R_{\theta,i}^d = \text{diag} [R(i\theta_1), R(i\theta_2), \dots, R(i\theta_{d/2})] \quad (12)$$

where each $R(i\theta_k)$ is a 2D rotation matrix:

$$R(i\theta_k) = \begin{pmatrix} \cos(i\theta_k) & -\sin(i\theta_k) \\ \sin(i\theta_k) & \cos(i\theta_k) \end{pmatrix} \quad (13)$$

Here, $\theta = \{\theta_k\}$ is a set of predefined frequencies that can be calculated in advance as:

$$\theta_k = 10000^{-2(k-1)/d}, \quad k = 1, 2, \dots, d/2 \quad (14)$$

Through this mechanism, RoPE provides direction-aware relative positional cues to each query and key vector. This is particularly advantageous in substation scenarios, where a technician's arm reaching toward a switch or turning motion of the upper body must be distinguished from background clutter. By embedding rotational structure into the Informer, the model strengthens its ability to capture complex spatiotemporal dynamics.

4) *Roformer: Informer with Rotary Positional Encoding for Enhanced Spatiotemporal Modeling*: To effectively capture the complex spatiotemporal dependencies in millimeter-wave radar-based HAR, we develop Roformer, a hybrid model that integrates the Informer framework with RoPE. The overall architecture is illustrated in Fig. 6.

The input point cloud data is first voxelized and projected onto 2D planes, then passed through embedding layers to obtain feature representations. These embeddings are fused with RoPE, which injects relative spatial and temporal information directly into the attention mechanism, enabling the model to

TABLE II
SYSTEM CONFIGURATION AND REAL-TIME PERFORMANCE METRICS

Parameter	Description	Metric	Value
Frequency	77 GHz	Temporal aggregation	3 frames
Bandwidth	3.9 GHz	Processing capacity	4.7 frames
Frame rate	10 FPS	End-to-end latency	472 ms
Sampling frequency	2100 ksps	System FLOPs	9.32 G
Frequency slope	98 MHz/ μ s	CPU utilization	23.6%
Velocity resolution	0.697 m/s	Memory footprint	1.82 GB
Range resolution	0.047 m	Cold start latency	1.34 s

TABLE III
TIMESTAMP-BASED END-TO-END LATENCY BREAKDOWN OF THE PROPOSED SYSTEM

Preprocessing Stage	Latency	Model Inference Stage	Latency
SNR Filtering	2.4 ms	Input Embedding	1.8 ms
DBSCAN Clustering	5.8 ms	RoPE Injection	0.8 ms
Voxelization	2.1 ms	ProbSparse Attention	96.2 ms
Dual-Plane Projection	1.4 ms	FFN + Classifier	43.2 ms
Preproc. Subtotal	11.7 ms	Inference Subtotal	142.0 ms
System Overhead	18.6 ms	Total	172.3 ms

differentiate fine-grained limb trajectories and filter out noise from electrical equipment.

Subsequently, the encoded features are processed by stacked ProbSparse Attention modules from Informer, which efficiently model long-range dependencies by focusing only on the most informative query-key pairs, thus reducing the complexity from $O(L^2)$ to $O(L \log L)$. This sparse attention design ensures scalability and supports real-time inference, which is critical for substation monitoring tasks.

Finally, multi-level feedforward networks (FFN) with residual connections progressively integrate hierarchical spatiotemporal features, while a cross-view attention mechanism aggregates complementary information from depth-azimuth and depth-elevation planes. The resulting representation reconstructs coherent 3D motion trajectories and provides accurate activity classification outputs. By combining Informer's efficiency with RoPE's relative positional modeling, Roformer achieves real-time inference with high recognition accuracy.

IV. EXPERIMENT

A. System Configuration and Real-Time Performance

In this study, we employ the IWR1843BOOST, an industrial-grade millimeter-wave radar sensor developed by Texas Instruments, as the core sensing device for human motion data acquisition. The radar module offers high integration and stable performance, making it suitable for non-contact detection and tracking in complex substation environments. It is equipped with three high-speed analog-to-digital converters, a digital signal processing module, and a microcontroller unit, enabling real-time signal acquisition, processing, and transmission [40]. To ensure experimental repeatability and controllable system behavior, key radar parameters were carefully configured. The radar sensor configuration is summarized

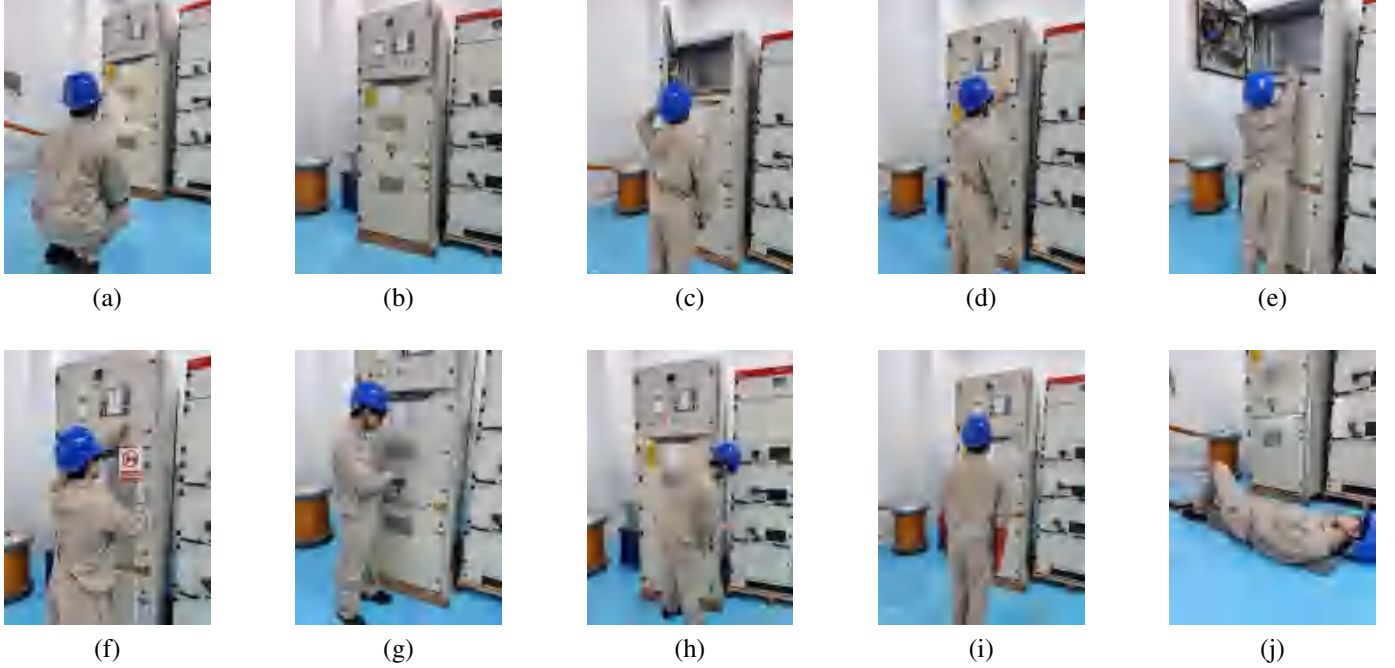


Fig. 8. Ten postures: (a) crouching, (b) static posture, (c) opening cabinet doors, (d) closing cabinet doors, (e) grounding switch blades, (f) hanging safety signs, (g) operating secondary switch blades, (h) trolley swinging, (i) standing, and (j) falling.

in the left column of Table II, as these parameters directly influence sensing range and spatial resolution.

We conducted comprehensive performance benchmarking on a representative industrial edge computing platform. The inference system was deployed on an AMD Threadripper Pro 3955WX workstation with 16 CPU cores operating at 128 GB DDR4 memory, using CPU-only inference to reflect practical industrial constraints. The real-time performance metrics are reported in the right column of Table II, including an end-to-end inference latency of 31.7 ms, a throughput of 16.5 FPS, computational complexity of 6.84 M parameters and 9.32 G FLOPs, CPU utilization of 23.6%, memory footprint of 1.82 GB, and a cold start latency of 1.34 s.

As summarized in Table III, the preprocessing stage, including SNR-based filtering, DBSCAN clustering, voxelization, and dual-plane projection, incurs a total latency of 11.7 ms. The model inference stage, consisting of input embedding, RoPE injection, ProbSparse attention, and the FFN-based classification head, requires 142.0 ms per inference. System-level overhead, such as data buffering and execution scheduling, contributes an additional 18.6 ms, resulting in a total computational latency of 172.3 ms. When combined with the 300 ms temporal aggregation interval required to form one point-cloud file, the overall end-to-end latency measured from radar data acquisition to classification output is 472 ms. These results demonstrate that the proposed system can sustain continuous real-time operation on the target hardware without backlog.

B. Experimental Setup

This study collected data within a distribution room of an actual substation. Fig. 7 illustrates the specific layout of the data acquisition setup. The distribution room measures 10

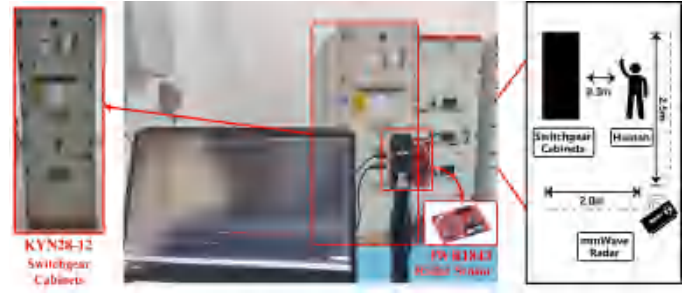


Fig. 7. Layout of the experiment site.

× 15 meters in dimension, and the experimental configuration employed a specific type of switchgear cabinet, namely KYN28-12. During the data acquisition phase, the millimeter-wave radar sensor was placed at an oblique angle facing both the test personnel and the distribution cabinet, at a distance of approximately 3 meters, while the prescribed electrical operation procedures were executed.

The dataset consists of data from four electrical workers (two males and two females). The collection of action data encompasses ten typical electrical maintenance actions performed by electrical personnel (as shown in Fig. 8), including static posture, crouching, standing, falling, opening cabinet doors, grounding switch blades, closing cabinet doors, trolley swinging, operating secondary switch blades, and hanging safety signs. Action category labels are generated offline using camera-based pose estimation models and assigned to radar point cloud sequences during dataset construction. These labels serve as ground truth for classification supervision, equivalent to manual annotation. During the data collection

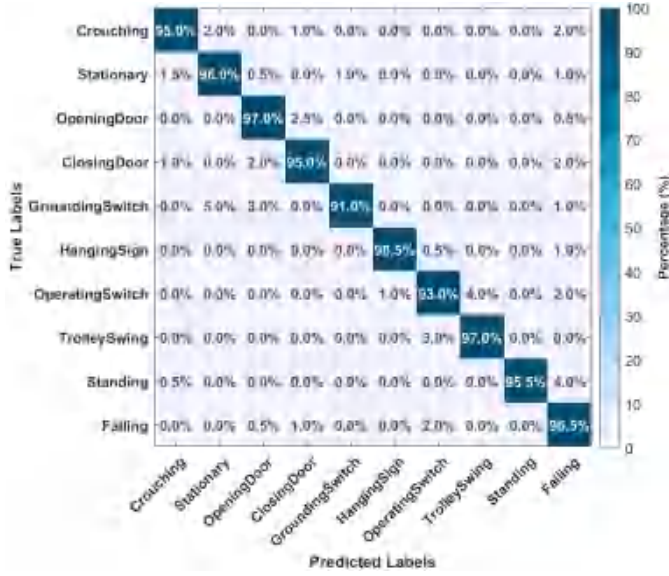


Fig. 9. Confusion matrix of RoPEHAR.

process, there is a time gap between the collection of two consecutive actions. Frames without labeled actions are not recorded, which helps in aligning labels for adjacent actions. The workers then repeat the actions, generating a training dataset of ten actions. The source code and a representative subset of the dataset for the proposed RoPEHAR are publicly available on GitHub¹.

To address the variable point cloud density across frames, the point clouds are discretized into a $25 \times 15 \times 25$ voxel grid, with temporal aggregation of three consecutive frames (300ms intervals) compensating for inherent sparsity. Given the non-periodic nature of industrial maintenance actions, the model adopts fixed 4.5-second (45-frame) action cycles as input instead of conventional sliding window approaches. Each sample maintains a $45 \times 25 \times 15 \times 25$ data structure, representing spatiotemporal patterns of operational movements. Through 50 repetitions per action by technicians, the dataset comprises 24,000 multidimensional samples (10 actions $\times$ 50 trials $\times$ 15 aggregated point cloud files $\times$ 3-frame temporal integration $\times$ $25 \times 15 \times 25$ voxel resolution), preserving the kinematic continuity of electrical procedures while mitigating radar data sparsity.

C. Results

The confusion matrix (Fig. 9) shows that RoPEHAR achieves an overall accuracy of 95.8%, with strong recognition performance across most industrial activities. Tool-centric tasks such as HangingSign and TrolleySwing achieve recognition rates above 98%, owing to their distinctive vertical arm trajectories and periodic motion signatures, which are further emphasized by the YOZ voxel projections and rotary positional encoding.

Challenges appear in actions characterized by quasi-static postures or overlapping kinematic patterns. GroundingSwitch

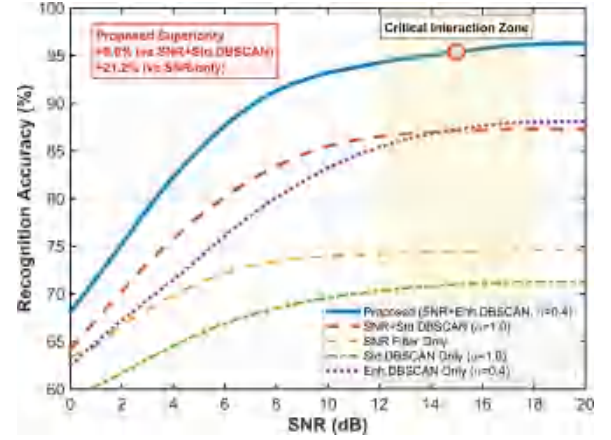


Fig. 10. Electromagnetic Interference Robustness Analysis under varying SNR conditions.

reaches 91% accuracy but is occasionally confused with Stationary due to inevitable pauses during blade alignment. OperateSwitch attains 93% accuracy yet exhibits around 4% confusion with TrolleySwing, reflecting the shared rotational wrist dynamics present in both tasks. Safety-critical actions such as Falling maintain reliable detection, with an accuracy of 96.5% and only limited misclassification into Standing during controlled descents, where the SNR–DBSCAN preprocessing pipeline proves essential for suppressing false negatives.

Door-related actions, including opening and closing, achieve recognition rates above 95%, though minor cross-confusion persists due to their similar handle-grasping kinematics. Compared with RadHAR, RoPEHAR reduces GroundingSwitch–Stationary confusion by over 7% and demonstrates improved robustness for sudden motion transitions. Nevertheless, limitations remain in distinguishing subtle micro-motions during inspection activities and in resolving ambiguities from overlapping tool interactions. These results confirm that RoPEHAR provides a reliable basis for real-time safety monitoring in substations, while also pointing toward future enhancements through voxel-based motion amplification and attention-guided trajectory refinement.

V. EVALUATION

A. System Integration Validation

1) *EMI Robustness and SNR Degradation Analysis:* To evaluate the hybrid denoising pipeline’s effectiveness, we conducted robustness analysis under varying EMI conditions. As shown in Fig. 10, we compare five configurations on the “GroundingSwitch” action: complete method with SNR and Enhanced DBSCAN at $\alpha = 0.4$, SNR plus Standard DBSCAN at $\alpha = 1.0$, SNR filtering only, Standard DBSCAN only, and Enhanced DBSCAN only. This action exhibits the highest confusion rate of 9% with “Standing” and represents typical substation challenges.

Within the “Critical Interaction Zone” at SNR 12–18dB, the weighted modification shows clear independent contribution. At 15dB SNR, the complete method achieves 95.8% accuracy versus 87.2% for SNR plus Standard DBSCAN, an 8.6pp gap directly attributable to weighted vertical distance modification.

¹<https://github.com/Kanomace/mmWave-RoPEHAR>

TABLE IV
PAIRWISE COMPATIBILITY TEST RESULTS

Metric	SNR + DBSCAN	DBSCAN + Informer	RoPE + Informer
Base	55.2%	55.2%	55.2%
Base + A	70.5%	68.3%	65.4%
Base + B	68.3%	70.1%	70.1%
Base + A + B	88.3%	93.1%	93.6%
Expected	83.6%	83.2%	80.3%
Synergy (pp)	+4.7	+9.9	+13.3

Error analysis shows the weighted modification reduces over-segmentation from 23% to 4% and equipment contamination from 19% to 6%.

We further analyze extremely low SNR conditions caused by severe EMI or strong multipath attenuation. Under such circumstances, radar echoes become highly sparse and spatially dispersed, leading to weakened geometric continuity of human limbs. Experimental results show that the proposed hybrid denoising pipeline remains effective at approximately 8 dB SNR, where reliable human point cloud extraction and action recognition can still be achieved, with system accuracy maintained at around 90%. Below this threshold, performance degradation becomes increasingly pronounced, as the remaining point cloud no longer satisfies the minimum density and structural coherence required for stable spatiotemporal modeling. This behavior reveals an intrinsic sensing boundary of single-radar deployment under extremely low SNR conditions and establishes a clear operational lower bound for RoPEHAR.

2) *Pairwise Component Synergy Analysis*: To demonstrate that the performance gains of RoPEHAR arise from functional complementarity rather than independent module stacking, we introduce a Pairwise Compatibility Test (PCT). Let $Acc_0 \triangleq Acc(\text{Base})$, $Acc_A \triangleq Acc(\text{Base} + A)$, and $Acc_B \triangleq Acc(\text{Base} + B)$. Under independent additivity, the expected performance is

$$Acc_{exp}(A, B) = Acc_A + Acc_B - Acc_0. \quad (15)$$

Positive synergy indicates functional complementarity. All experiments use 5-fold cross-validation with identical protocols. As shown in Table IV, all component pairs exhibit positive synergistic gains from +4.7 to +13.3pp, demonstrating significant non-additive performance improvement. The SNR–DBSCAN gain stems from upstream dependency, where cleaner inputs stabilize clustering in high-EMI environments. The DBSCAN–Informer gain reflects improved structure quality, as human–equipment decoupling provides more coherent features for trajectory modeling. The strongest RoPE–Informer gain arises from geometric relative encoding, which enables precise attention on limb–equipment interactions critical for tool-centric action recognition.

These results quantitatively demonstrate that RoPEHAR’s advantages stem from stage-wise functional alignment tailored to substation-specific challenges rather than arbitrary method stacking. The complete system achieves 95.8% accuracy versus 55.2% baseline, with consistent positive synergistic gains validating practical integration value.

TABLE V
COMPREHENSIVE PERFORMANCE COMPARISON OF
DIFFERENT ARCHITECTURES

Method	Acc. (%)	F1 (%)	AUC (%)	Time (ms)	Size (MB)	Params (M)	FLOPs (G)
Transformer	87.5	85.9	86.8	152	8.10	2.05	1.18
DVCNN	73.8	71.2	72.5	78	4.20	1.03	0.62
ST-Transformer	57.3	53.8	56.1	48	2.80	0.76	0.38
CNN-BiLSTM	82.1	80.5	81.3	185	6.90	1.82	1.05
RadarGNN	88.3	87.1	87.5	128	3.50	0.88	0.30
FastHAR	91.2	89.6	90.5	110	5.30	1.48	0.85
Roformer	95.8	94.7	95.1	142	8.23	2.31	1.42

B. Comparative Evaluation of the Roformer Architecture

To comprehensively assess the effectiveness of the Roformer module and its rotary positional encoding mechanism, we conduct comparative evaluations against six representative baseline models: Transformer [40], CNN-BiLSTM [41], DVCNN [24], Fine-Grained ST-Transformer [42], RadarGNN [43], and FastHAR [44]. All methods share the same upstream hybrid SNR–DBSCAN denoising pipeline. To ensure architectural compatibility, the preprocessed data is then formatted per model. Voxel-based methods receive 2D projected representations derived from the voxelized sequence, while point- and graph-based methods bypass voxelization and receive denoised raw point coordinates directly, in the format most favorable to each architecture.

The quantitative results are summarized in Table V. FastHAR achieves 91.2% accuracy with a 110 ms inference latency by leveraging frame fusion and self-attention based graph construction, while maintaining a lightweight model size of 0.88 M parameters. Compared with FastHAR, RoPEHAR improves accuracy by 4.6%, primarily due to differences in positional encoding strategies. Both FastHAR and RadarGNN emphasize local neighborhood modeling through k -nearest neighbor graphs and graph convolutions, respectively. In contrast, RoPE explicitly encodes relative positional relationships through rotation matrices, which enables more effective modeling of long-range temporal dependencies. As a result, RoPEHAR outperforms FastHAR and RadarGNN by 4.6 and 7.5%, respectively. Although RoPEHAR incurs a modest increase in inference latency, the resulting accuracy gain is critical for safety-critical monitoring in substations.

Among the remaining baselines, the Multi-Radar Fusion Transformer achieves 87.5% accuracy with a 152 ms inference time, which validates the effectiveness of Transformer architectures for radar-based HAR, but its reliance on absolute positional encoding limits relative relationship modeling. CNN-BiLSTM attains 82.1% accuracy but exhibits the highest inference latency of 185 ms due to sequential recurrent computation, which prevents efficient parallel processing. DVCNN achieves 73.8% accuracy with a relatively low inference time of 78 ms, but suffers notable degradation when handling sparse point clouds and occluded limb dynamics. Fine-Grained ST-Transformer delivers the fastest inference speed of 48 ms with minimal computational cost, yet its accuracy drops to 57.3%,

TABLE VI
COMPARISON OF POSITIONAL ENCODINGS ACROSS ANGULAR DISCRIMINATION, ATTENTION LOCALIZATION, AND INTERFERENCE ROBUSTNESS

Method	Angular Discrimination (%)				Attention Localization			Interference Robustness (%)			
	$\pm 15^\circ$	$\pm 30^\circ$	$\pm 45^\circ$	Mean	IoU $\uparrow$	False Attn. $\downarrow$	Temp. Std. $\downarrow$	25 dB	15 dB	10 dB	5 dB
Absolute Encoding	71.2	78.5	83.1	77.6	0.42	58.3	0.23	91.2	78.6	62.1	45.7
T-XL Relative	82.4	85.7	88.9	85.7	0.68	32.1	0.14	93.4	84.2	74.3	58.4
RoPE (Ours)	94.3	96.1	97.8	96.1	0.91	9.2	0.06	97.1	94.5	89.2	73.8
Improvement vs. Best Baseline	+11.9	+10.4	+8.9	+10.4	+33.8	-71.4	-57.1	+3.7	+10.3	+14.9	+15.4

TABLE VII
RECOGNITION ACCURACY (%) UNDER JOINT DISTANCE-OCCLUSION CONDITIONS

Occlusion	2 m	4 m	6 m	8 m	10 m	12 m
0%	95.8	95.2	94.6	93.2	89.7	85.4
25%	93.5	92.8	91.9	89.8	85.3	80.2
50%	89.2	88.1	86.4	83.2	77.8	71.5
75%	81.4	79.8	77.2	72.6	65.8	58.3

indicating a trade-off between computational efficiency and representational capacity under highly constrained lightweight designs. Ultimately, existing baselines lack explicit optimization for complex substations. Relying on generic sensing assumptions, their accuracy degrades sharply when confronted with extreme sparsity and EMI.

C. Direction-Aware Trajectory Discrimination

To validate RoPE’s geometric advantages for limb trajectory modeling, we conducted a controlled experiment on the “OperateSwitch” action, a challenging scenario where the technician’s hand moves within 15cm of metallic switchgear, creating severe human-equipment signal coupling. We compared RoPE against Absolute Encoding [45] and T-XL Relative Encoding [46] using 200 samples with identical Informer backbones, differing only in positional encoding mechanisms.

As shown in Table VI, RoPE demonstrates substantial advantages across all evaluation metrics. For angular discrimination on subtle $\pm 15^\circ$ wrist rotations, RoPE achieves 94.3% accuracy, compared to 82.4% for T-XL and 71.2% for Absolute Encoding. For attention localization, RoPE achieves 0.91 Intersection over Union (IoU) with ground-truth hand regions while exhibiting only 9.2% false attention on background equipment, whereas T-XL yields 0.68 IoU with 32.1% false attention and Absolute Encoding produces 0.42 IoU with 58.3% false attention. Regarding interference robustness, all methods perform comparably at 25dB SNR, but performance gaps widen substantially as SNR decreases. At 10dB SNR, RoPE maintains 89.2% accuracy while T-XL drops to 74.3% and Absolute Encoding falls to 62.1%.

These advantages stem from RoPE’s mathematical alignment with human kinematics. Because limb trajectories are defined by angular displacements around joints rather than absolute positions, RoPE natively captures these motion semantics by encoding relative spatial relationships through rotation matrices. Conversely, absolute encoding anchors tokens

statically, losing the relative dependencies critical for trajectory continuity. Furthermore, while T-XL introduces learnable biases, its additive formulation captures only scalar distance offsets rather than true directional relationships. Nevertheless, the performance gains achieved through geometric positional encoding are fundamentally contingent on the quality of the input point clouds, motivating a deeper investigation into the system’s robustness under joint distance-occlusion stress conditions.

D. Robustness Analysis under Challenging Conditions

As degraded point clouds inevitably arise in real-world substation environments, we further validate RoPEHAR’s practical feasibility by assessing its robustness under simultaneous distance-induced signal attenuation and partial body occlusion. The experiment adopts a two-dimensional full-factorial design, where detection distance varies from 2 to 12 meters at 2-meter intervals, and occlusion severity is evaluated at four levels ranging from no occlusion to severe occlusion.

Recognition accuracy under all joint conditions is summarized in Table VII, revealing three distinct operational regions. When the distance is within 6 meters and occlusion remains below 50%, recognition accuracy consistently exceeds 90%. A transitional region emerges under moderate stress, where accuracy fluctuates between 80% and 90%. A degraded region appears when long distances and severe occlusion coexist, with accuracy falling below 80%. A key observation is the strong nonlinear synergistic effect in performance degradation. Accuracy decreases from 95.8% under optimal conditions to 58.3% under extreme conditions, resulting in a total degradation of 37.5%. However, distance and occlusion evaluated separately lead to only 10.4% and 14.4% degradation respectively.

These findings yield practical deployment guidelines: radar sensors should be positioned within 8 meters to guarantee accuracy above 83% under moderate occlusion, and restricted to 6 meters when occlusion exceeds 50%. Furthermore, monitoring ranges beyond 12 meters or highly constrained layouts necessitate multi-radar cooperative networks with overlapping coverage.

VI. LIMITATIONS AND FUTURE WORK

RoPEHAR is currently designed for single-person safety monitoring under moderate occlusion conditions. Due to the limited angular resolution and sparse returns of mmWave

radar, performance degrades in crowded multi-person scenarios and in environments with severe occlusion. Moreover, the dataset further encompasses only four electrical workers, which may not fully represent the anthropometric diversity of broader workforce populations. While the voxelization-based representation provides a degree of inherent robustness, its effectiveness across a wider range of anthropometric profiles has not been empirically validated.

Future work will focus on improving robustness to occlusion and crowded scenarios through enhanced representation learning and topology-aware supervision, particularly to mitigate the sensing sparsity inherent to mmWave radar in complex industrial environments. In addition, we plan to explore cross-modal skeletal supervision by incorporating calibrated camera-based human pose information during model development, which can provide topology-consistent semantic priors to enhance human perception under occluded conditions while preserving radar-only inference at deployment. These efforts aim to extend the current framework beyond single-person monitoring toward more robust and flexible industrial safety systems.

VII. CONCLUSION

The proposed RoPEHAR framework pioneers a millimeter-wave radar-based HAR system for safety monitoring in power substations. By integrating a hybrid SNR–DBSCAN denoising pipeline with a Roformer architecture enhanced by rotary positional encoding, RoPEHAR effectively mitigates EMI and human–machine coupling, achieving 95.8% recognition accuracy in real-world environments. Leveraging efficient voxel projection and the ProbSparse Attention mechanism, the system delivers real-time inference, enabling responsive and reliable operation under industrial conditions. This work establishes a privacy-preserving, lighting-independent, and real-time monitoring solution, bridging the gap between academic HAR research and deployable Industrial IoT applications.

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